Skewness

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Introduction

Skewness is the third standardised moment of a distribution; it quantifies how asymmetric the distribution is around its mean. Zero skewness corresponds to perfect symmetry; positive skewness indicates a long right tail; negative skewness indicates a long left tail. Many biomedical variables (lab values, incubation times, survival times) are right-skewed; income and wealth distributions are classically right-skewed with very long tails.

Prerequisites

The reader should know what a mean and SD are, and should have seen a few histograms.

Theory

The population skewness is

$$\gamma_1 = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right].$$

Several sample estimators exist with different small-sample biases. The classical moment estimator is $g_1 = m_3/m_2^{3/2}$ where $m_k = \frac{1}{n}\sum(x_i - \bar{x})^k$. SAS/SPSS/`psych` use a bias-corrected form $G_1 = \sqrt{n(n-1)}/(n-2) \cdot g_1$. Interpretation: $|g_1| < 0.5$ is approximately symmetric, $0.5 < |g_1| < 1$ is moderately skewed, $|g_1| > 1$ is highly skewed.

Assumptions

As a descriptive statistic, skewness has no assumptions. As an estimator, it requires finite third moments, which fails for heavy-tailed distributions like the Cauchy.

R Implementation

```r
library(moments)

set.seed(2026)
sym   <- rnorm(2000, 0, 1)
rskew <- rlnorm(2000, meanlog = 0, sdlog = 0.7)
lskew <- -rlnorm(2000, meanlog = 0, sdlog = 0.7)

c(symmetric = skewness(sym),
  right     = skewness(rskew),
  left      = skewness(lskew))

# log transform reduces right skewness
skewness(log(rskew))
```

Output & Results

```
symmetric      right       left
    0.013      2.134     -2.090
```

Log-transforming the right-skewed sample drops skewness from 2.1 to approximately 0; this is the standard remedy for biomedical lab values.

Interpretation

Report skewness for any variable whose histogram departs from symmetry. When $|g_1| > 1$, the mean is pulled away from the centre of mass; the median is a better point summary, or the variable should be transformed.

Practical Tips

- Log transformation is the default remedy for right-skewed positive data.
- Visual inspection beats numerical thresholds; a histogram makes the shape immediately visible.
- Skewness in small samples is noisy; avoid thresholds below $n = 30$.
- The Pearson skewness coefficient 3(mean − median)/SD is a robust alternative.
- For formal tests of skewness, use D'Agostino's test (`moments::agostino.test`).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1